Soal 7 – Reliability of the E-car Product Attribute (TB) Scale

Oct 12, 2026 · @azka

Summary

The six TB items do not form an internally consistent scale in this dataset: Cronbach's alpha = 0.432 (95% CI 0.353–0.506) and the odd-even split-half reliability corrected by Spearman–Brown = 0.507, both well below the 0.70 benchmark. Only one of six items (TB4) reaches a corrected item–total correlation of 0.30.

The analysis uses all 510 respondents in the published file. Removing one of two rows that are identical on all 41 variables (n = 509) changes no result by more than 0.004.

The items were chosen on content before any statistic was computed, and no item was dropped afterwards. The low reliability is therefore a finding about this item set, not an artefact of item selection.

The official Soal 7 instruction text was not available to the analyst. The criteria used (CITC ⩾ 0.30; reliability ⩾ 0.70) are stated assumptions and may need to be aligned with the lecturer's criteria. The compliance check below lists what can and cannot yet be confirmed.

Assignment Instructions and Compliance Check

The official Soal 7 (Question 7) text has not been found, so it is not reproduced here and full compliance cannot yet be claimed. The matrix below is checked against an unofficial summary of the requirements; every "Met" must be re-confirmed against the official text.

Where the official text was searched (12 Oct 2026): the analysis repository (all 18 remote branches); all files uploaded to this session; the session's working folders; the student's Google Drive (keyword and recent-file searches); and the student's published Claude artifacts. None contains the official exam text.

Unofficial sources of the requirements used below: (1) the student's own summary of the instructions given during the analysis, and (2) a requirements checklist that paraphrases the exam, found in another document. Neither is the exam wording. The second suggests that Question 7 also asks for a construct linked to the exam's case context (energy prices / EV transition, an assigned country and recent news), content validity and target respondents.

Requirement (unofficial paraphrase)	Status	Where addressed / gap
Six Likert items measuring one coherent construct	Met, with caveat	TB1–TB6 chosen on content (Construct and Item Selection). The data show the six items do not behave as one construct (alpha 0.432).
Content validity of the items	Partly met	Only the analyst's wording review. No expert panel or content validity index was carried out.
Target respondents defined	Partly met	Respondents appear to be Indian students (inferred from SPSS labels). Why they are the right target group for the exam's case is not argued.
Public item-level dataset with at least 30 respondents (or own data from at least 30 consenting participants)	Met, pending licence check	n = 510 item-level responses. Licence, version and citation still unverified.
Corrected item–total correlations	Met	Results; workbook sheet Item_Analysis
Odd-even split-half with Spearman–Brown correction	Met	Results; workbook sheet Reliability
Cronbach's alpha	Met	Results; workbook sheet Reliability
Link to the exam's case context (topic, assigned country, recent news)	Cannot assess / likely gap	The dataset concerns electric-car perceptions of students in India. Whether this fits the required topic and the student's assigned country is unknown without the official text.
English report; α = 0.05	Met	Whole report; p-values at α = 0.05

Requirement (unofficial paraphrase)	Status	Where addressed / gap
Dataset cited with URL, code, extraction date and period	Partly met	URL and Mendeley ID given. DOI, licence, download date and data-collection period still missing.
Original data included	Met	Workbook sheet Raw_Data holds all 510 rows unchanged
File naming (StudentID_Name_UTS)	Not assessed	Applies to the final submission file
AI use disclosed with prompts and verification steps	Met (to be adapted)	AI Use Disclosure section
Protocol-only option (maximum 6/12)	Not applicable	Used only if no qualifying dataset is available; a dataset was used

Gaps to close once the official text is available: a content-validity step for the six items; a justification of the target respondents; the link between the construct and the exam's case (including the assigned country and any recent-news requirement); and the full dataset citation.

Data Source and Provenance

The raw files are verified; the dataset's licence, version, citation and scale anchors are not. Each claim below is labelled by how it was established.

Item	Status	Evidence
510 respondents × 41 variables, one row per respondent	Verified	Uploaded CSV and SPSS files, read directly
31 Likert-type items coded 1–5, no missing values	Verified	15,810 item cells checked in both files
CSV and SPSS item values identical	Verified	Cell-by-cell comparison, 0 differences
Item wording	Verified	SPSS variable labels
SPSS file created 21 May 2021 (SPSS 23)	Verified	SPSS file header

Item	Status	Evidence
Dataset page: Mendeley Data, "Electric Car Purchase Intention", version 2 (link)	Not verified	Supplied by the student; the page could not be opened from the analysis environment
Licence CC BY 4.0	Not verified	Stated by the student; must be confirmed on the dataset page
Authors	Inference	The SPSS file name contains "Maheswaran_and_Veena"; a related study by Veena Pailwar and Maheswaran Srinivasan (IMT Nagpur) was found through search results only (European Transport 89, 2022; Academy of Marketing Studies Journal 26(S2), 2022)
Link between this file and those articles	Not verified	Search snippets report samples of 436 and 273 students, not 510; neither article could be opened
Meaning of codes 1–5	Not verified	The SPSS file has no value labels for the items
Name of the TB construct	Not verified	Only the abbreviation "TB" appears in the files

Attribution (to be completed from the dataset page): [Authors as listed on the page] ([year]). *Electric Car Purchase Intention* (Version 2) [Data set]. Mendeley Data. https://doi.org/10.17632/tz7krp8rvz.2. Licensed under CC BY 4.0 [confirm]. The DOI follows Mendeley's standard pattern (prefix 10.17632 plus dataset ID and version) and must be checked on the page.

Related publications (from search results only; not opened, not verified):

- Pailwar, V. K., & Srinivasan, M. (2022). Indian management students' electric car purchase intention. *European Transport | Trasporti Europei*, 89, Paper 7. https://doi.org/10.48295/ET.2022.89.7
- Pailwar, V. K., & Srinivasan, M. (2022). Management students' electric car purchase intentions: An exploratory study. *Academy of Marketing Studies Journal*, 26(S2). Issue and page details differ between search results and must be checked.

Sample-size discrepancy. The dataset has 510 rows. Search snippets attribute 436 respondents to the European Transport paper and, in one snippet, 273 valid responses to the exploratory paper; another snippet attributes 436 to the exploratory paper. The reason for the difference cannot be established from the available material. Plausible

explanations include screening or exclusion of responses before the published analyses, data collected in more than one round, or a deposit that does not correspond exactly to either paper. None of these is assumed in this report. The 510-row file is analysed as published, and it should not be described as the sample of either article until the dataset page or the articles confirm the link.

Construct and Item Selection

TB1–TB6 were selected because every item rates an attribute of the electric car itself, with the same stem ("I think e-cars are…"). This judgement was made from wording alone, before any correlation or reliability statistic was computed.

Order	Column	Wording (SPSS label)	Facet (analyst's reading)
1	TB1	I think e-cars are sleek	Aesthetic
2	TB2	I think e-cars are durable	Functional
3	TB3	I think e-cars are available in different size, colours and designs	Aesthetic / variety
4	TB4	I think e-cars are aesthetically appealing	Aesthetic
5	TB5	I think e-cars are technologically superior	Functional
6	TB6	I think e-cars give comfortable ride	Functional

In this report the construct is called **perceived e-car product attributes**. This label is the analyst's description; the original construct name is not documented in the files. Search snippets say the related study found a student preference for "automotive design", which may correspond to TB, but this could not be confirmed.

The only other group with exactly six items, DC1–DC6, was rejected on content. It mixes external infrastructure (charging points on highways, service centres) with vehicle performance (charging speed, range, pickup) and noise. DC was not analysed afterwards, so the scale choice could not be steered by the results.

Rules fixed before computing:

- Item order 1–6 follows the SPSS variable order (TB1…TB6).
- Odd half = TB1 + TB3 + TB5; even half = TB2 + TB4 + TB6.
- No item is dropped after seeing results; alpha-if-item-deleted is reported for diagnosis only.
- Main analysis on n = 510; sensitivity analysis on n = 509.

Data Preparation

No data were changed, imputed or recoded; all six TB items are complete for all 510 respondents.

Check	Result
Missing values, TB1–TB6	0 of 3,060 cells
Values outside 1–5	0
Negatively worded items needing reverse coding	None: all six are positive statements
Rows identical on all 41 variables	2 rows (data rows 138 and 141)
Implausible values outside the TB items	Age = 2 in data row 347; inconsistent coding of `Comtd_E_car` (0/1/blank, labelled 1 = Yes, 2 = No); broken value labels for the three ownership variables. None of these enter the analysis.

Duplicate rows. Two rows match on all 41 variables, including 31 Likert answers. That pattern is unlikely by chance, so double entry is plausible, but two students could also have answered identically. The rows are therefore not treated as an error. The main analysis keeps both (n = 510, as published), and a sensitivity analysis removes the second occurrence (n = 509).

Scale coding. The files do not state what codes 1–5 mean. Correlations, corrected item–total correlations, split-half reliability and alpha do not change if every item is reversed in the same way, so the reliability results hold either way. Item means, however, are interpreted below only under the assumption that higher codes mean stronger agreement. That assumption is consistent with the agreement-type wording but is not documented.

Methods

Three statistics were computed on the six TB items, with sample variances (n − 1) throughout, matching SPSS and Excel's `VAR.S`.

1. Corrected item–total correlation (CITC). The Pearson correlation between each item and the sum of the other five items (the "rest score"). Leaving the item out of its own total prevents the inflation seen in the uncorrected correlation.

```
r_{i,\,\text{rest}} = \operatorname{corr}\left(X_i,\; \sum_{j \neq i} X_j\right)
```

2. Odd-even split-half reliability with Spearman–Brown correction. Half scores are Odd = TB1 + TB3 + TB5 and Even = TB2 + TB4 + TB6. Their Pearson correlation r_hh estimates the reliability of a three-item half; Spearman–Brown projects it to six items.

```
r_{SB} = \frac{2\,r_{hh}}{1 + r_{hh}}
```

3. Cronbach's alpha. With k = 6 items, item variances s^2_i and total-score variance s^2_T:

```
\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} s_i^2}{s_T^2}\right)
```

A 95% confidence interval for alpha uses Feldt's F-based method, with $df_1 = n - 1$ and $df_2 = (n - 1)(k - 1)$. Alpha-if-item-deleted and the inter-item correlation matrix are reported for diagnosis.

Criterion (assumed)	Threshold	Basis
Item discrimination	CITC ⩾ 0.30	Widely used rule of thumb
Statistical significance of CITC	p < 0.05 (r_crit = 0.087 at n = 510)	Reported, but not used to judge items: with n = 510 even trivial correlations are significant
Reliability (alpha, Spearman–Brown)	⩾ 0.70 acceptable; 0.60–0.69 questionable; < 0.60 poor	Common convention

The lecturer's own thresholds should replace these if the instructions state different ones. In the workbook, the thresholds sit in yellow input cells and all decisions update automatically.

Results

Five of six items fall below CITC = 0.30, and both reliability coefficients are in the "poor" range (< 0.60).

Item statistics (main analysis, n = 510)

Item	Mean	SD	r uncorrected	CITC	p (CITC)	Alpha if deleted	CITC ⩾ 0.30?
TB1 sleek	3.755	0.973	0.592	0.292	< 0.001	0.335	No

Item	Mean	SD	r uncorrected	CITC	p (CITC)	Alpha if deleted	CITC ⩾ 0.30?
TB2 durable	3.069	1.013	0.509	0.171	< 0.001	0.414	No
TB3 size/colour/design variety	4.051	0.874	0.461	0.168	< 0.001	0.411	No
TB4 aesthetically appealing	3.339	0.801	0.564	0.321	< 0.001	0.331	Yes
TB5 technologically superior	2.906	0.938	0.412	0.087	0.051	0.460	No
TB6 comfortable ride	3.047	0.909	0.536	0.245	< 0.001	0.367	No

The uncorrected correlations (0.41–0.59) look acceptable, but they are inflated because each item makes up one-sixth of its own total. The corrected values (0.09–0.32) are the ones used for decisions.

Reliability (main analysis, n = 510)

Statistic	Value	Decision
Cronbach's alpha	0.432 (95% CI 0.353–0.506)	Poor
Standardised alpha (from mean inter-item r)	0.440	Poor
Odd-even half-score correlation r_hh	0.340	—
Spearman–Brown corrected split-half	0.507	Poor
Guttman split-half, same split (supplementary)	0.507	Poor
Mean inter-item correlation (15 pairs)	0.116 (range −0.145 to 0.309)	—

Deleting TB2, TB3 or TB5 would raise alpha only slightly (to 0.414, 0.411 and 0.460), and no deletion comes close to 0.70. No item removal would rescue the scale, and none was made. The weakest pair is TB3 (variety) with TB5 (technological superiority), r = −0.145. The strongest pair is TB1 (sleek) with TB4 (aesthetically appealing), r = 0.309.

Sensitivity analysis (n = 509, second identical row removed)

Statistic	n = 510	n = 509	Change
Cronbach's alpha	0.4325	0.4327	+0.0002
Spearman–Brown split-half	0.5073	0.5075	+0.0002
CITC range	0.087–0.321	0.090–0.320	⩽ 0.004 per item
Items with CITC ⩾ 0.30	1 (TB4)	1 (TB4)	none

The duplicate row has no material effect on any conclusion.

Interpretation

The six TB items do not measure one common attitude strongly enough to be summed into a single score. A student who rates e-cars as sleek is only weakly more likely to rate them as durable, comfortable or technologically superior (mean inter-item r = 0.116).

Three points matter for reading this result correctly:

- **Internal consistency is not validity.** CITC, split-half and alpha describe whether the items move together. A low alpha shows the items are not interchangeable indicators of one construct; it says nothing on its own about whether each item measures something meaningful. A high alpha would not have proved validity either.
- **The pattern fits the content concern raised before the analysis.** The two strongest correlations are among appearance items (TB1–TB4 r = 0.309; TB3–TB4 r = 0.276), while functional items correlate weakly with each other (0.08–0.16) and TB5 relates negatively to TB3. This suggests appearance and functional performance are separate judgements. It is an inference from a correlation pattern, not a factor-analytic result.
- **Sample size is not the cause.** With n = 510 the alpha interval is narrow (0.353–0.506). The low reliability is estimated precisely; it is not noise from a small sample.

The split-half coefficient (0.507) is higher than alpha (0.432). This is possible because alpha equals the average of all possible split-half coefficients computed with the Flanagan–Rulon (Guttman) formula, and any single split can land above or below that average. For this split the Guttman value (0.507) is almost identical to Spearman–Brown, because the two halves have similar variances (2.84 and 3.09). The fixed odd-even split is a pre-specified, not a favourable, choice.

Management Recommendations

The main recommendation concerns measurement: do not report or act on a single "product attribute" score built from TB1–TB6. The item-level points below assume that higher codes mean stronger agreement, which is not documented in the files.

1. **Track appearance and performance separately.** Marketing and product teams researching e-car perceptions should use separate, purpose-built multi-item scales for appearance and for functional performance (durability, comfort, technology). Each needs enough items (typically four or more), piloted and checked for reliability before use.
2. **Read the six items individually, not as an index.** In this sample, variety of sizes, colours and designs (TB3, mean 4.05) and sleekness (TB1, mean 3.75) received the most favourable ratings. Technological superiority (TB5, 2.91), comfort (TB6, 3.05) and durability (TB2, 3.07) sat near the scale midpoint.
3. **If those means reflect agreement, address functional doubts with evidence.** Test drives, battery and durability warranties, and independent reliability data target the attributes rated least favourably. Design-led messaging alone would emphasise attributes already rated well.
4. **Treat these as hypotheses for this segment only.** The SPSS labels describe respondents as students and record family income in INR, which suggests Indian students, but the sampling frame is not documented. The findings should be confirmed with target buyers before guiding investment.

Item means are descriptive. They were not compared by significance test, and no claim is made about purchase behaviour.

Limitations

The largest limitation is documentation: the construct name, the meaning of codes 1–5, the licence and the link to a published study could not be verified.

- **Undocumented scale anchors.** The reliability statistics are unaffected (they are invariant to reversing all items), but the item means in the recommendations depend on an assumed coding.
- **Undocumented construct.** "Perceived e-car product attributes" is the analyst's label. If the original authors did not design TB1–TB6 as one reflective scale, low internal consistency is less surprising and says less about the instrument.
- **Provenance gap.** The file has 510 rows, while the related articles reportedly analysed 436 and 273 students. The relationship between this dataset and those analyses is unknown.

- **Ordinal data.** Five-point responses are ordinal; Pearson-based CITC and alpha treat them as interval. A Spearman check gave similar CITCs (0.09–0.31). Polychoric (ordinal) alpha was not computed.
- **Alpha's assumptions.** Alpha assumes one underlying factor and equal item loadings. The correlation pattern suggests more than one factor, which also lowers alpha. Dimensionality was not tested with factor analysis.
- **One split only.** The odd-even split is one of ten possible ways to split six items into halves; another split would give a different coefficient.
- **Duplicate rows.** Two identical rows were kept in the main analysis; the sensitivity check shows their effect is negligible.
- **Assumed criteria.** The 0.30 and 0.70 thresholds are conventions, not the lecturer's stated criteria, which were unavailable.
- **Sample.** Respondents appear to be students, a convenience segment rather than the car-buying population.

Conclusion

The six TB items cannot be treated as a reliable single scale in this dataset (alpha = 0.432; Spearman–Brown split-half = 0.507; one of six items with CITC ⩾ 0.30). This result holds for both n = 510 and n = 509. It was not produced by item selection: the items were fixed on content beforehand and none were removed.

The evidence points to appearance and functional performance being separate judgements, which a revised instrument should measure with dedicated scales. Before submission, three things should be confirmed on the dataset page or in the original article: the licence and citation, the meaning of codes 1–5, and the construct name for TB.

AI Use Disclosure and Verification Log

An AI assistant (Claude by Anthropic, used through Claude Code) was used for data checking, calculation, building the workbook and drafting this report. The student chose the dataset, approved the item set and the duplicate-row policy, and is responsible for the submitted work. The wording of this section should be adapted to the format the lecturer requires.

What the AI did

- Read the uploaded CSV and SPSS files and ran the data checks.
- Proposed TB1–TB6 on content grounds and set the analysis rules before computing.
- Computed all statistics in Python (pandas, scipy) and rebuilt them as Excel formulas.

- Searched the web for the dataset page and source articles. Most repositories (Mendeley Data, publishers) were blocked from its environment, so only search-result snippets were seen; those are labelled "not verified" in this report.
- Drafted this report.

Verification performed

Check	Method	Result
CSV vs SPSS item values	Cell-by-cell comparison	0 differences in 15,810 cells
Workbook raw data vs CSV	Cell-by-cell comparison	0 differences
Excel formulas vs Python	60 statistics compared (Cross_Check sheet)	Largest absolute difference 7.5×10^{-15}
Duplicate detection	Excel formula and Python, independently	Both identify data rows 138 and 141
Formula errors	Full recalculation in LibreOffice	0 errors in 16,732 formulas

Final audit. Every number quoted in the Results tables was checked automatically against the workbook (script `audit_report.py`). Two rounding errors were found and corrected in this report: TB1's corrected item–total correlation (0.2925 rounds to 0.292, not 0.293) and TB1's alpha-if-deleted (0.3345 rounds to 0.335, not 0.334). The underlying statistics and all decisions were unchanged. The committed verification scripts were re-run from the repository and reproduced every reference value.

Main prompts given to the AI (summary, translated from Indonesian):

1. Check the uploaded dataset files and record what each contains; do not compute anything until all files are sent.
2. Carry out the validity and reliability analysis for Question 7 without inventing data, sources or results; verify every calculation.
3. Find and verify a public item-level dataset with at least 30 respondents and six Likert items measuring one construct; choose items by content, not by results.
4. Use TB1–TB6, with n = 510 as the main analysis and n = 509 as a sensitivity analysis; compute corrected item–total correlations, odd-even split-half with Spearman–Brown, and Cronbach's alpha; build a traceable workbook and an English report.
5. Audit the final work against the official Question 7 instructions, and do not claim full compliance until they are matched.

Not verified

- The workbook was recalculated in LibreOffice and has not been opened in Microsoft Excel.
- Licence, citation, version, scale anchors, construct name and the official Soal 7 criteria (see Data Source and Provenance).
- The student should independently re-check the key figures (for example by opening the workbook in Excel or running the same analysis in SPSS) before submission.

Appendix: Excel Formulas

All results come from formulas in `Soal7_TB_Reliability.xlsx`; they also work in Google Sheets. In Raw_Data, TB1–TB6 are columns AJ–AO, rows 2–511. TB_Primary links them into columns B–G; the sensitivity sheet has the same layout with 509 rows (rows 2–510).

Step	Cell	Formula
Link item from raw data	TB_Primary!B2	`=Raw_Data!AJ2`
Total score	TB_Primary!H2	`=SUM(B2:G2)`
Rest score for TB1	TB_Primary!I2	`=H2-B2`
Odd half	TB_Primary!O2	`=B2+D2+F2`
Even half	TB_Primary!P2	`=C2+E2+G2`
CITC, TB1	Item_Analysis!G10	`=CORREL(TB_Primary!$B$2:$B$511,TB_Primary!$I$2:$I$511)`
p for CITC	Item_Analysis!I10	`=TDIST(ABS(H10),$B$6-2,2)` with H10 = `G10*SQRT(($B$6-2)/(1-G10^2))`
Alpha if TB1 deleted	Item_Analysis!J10	`=(5/4)*(1-(SUM($E$10:$E$15)-E10)/VAR(TB_Primary!$I$2:$I$511))`
Sum of item variances	Reliability!B9	`=SUM(Item_Analysis!$E$10:$E$15)`
Variance of total	Reliability!B10	`=VAR(TB_Primary!$H$2:$H$511)`
Cronbach's alpha	Reliability!B11	`=(B8/(B8-1))*(1-B9/B10)`
Alpha 95% CI (Feldt)	Reliability!B12:B13	`=1-(1-B11)*FINV((1-$C$2)/2,B7-1,(B7-1)*(B8-1))` and `=1-(1-B11)*FINV(1-(1-$C$2)/2,B7-1,(B7-1)*(B8-1))`

Step	Cell	Formula
Half-score correlation	Reliability!B15	`=CORREL(TB_Primary!$O$2:$O$511,TB_Primary!$P$2:$P$511)`
Spearman–Brown	Reliability!B16	`=2*B15/(1+B15)`
Decision	Reliability!B14	`=IF(B11>=$C$3,"Acceptable",IF(B11>=$C$4,"Questionable","Poor"))`

`VAR`, `STDEV`, `TDIST`, `TINV` and `FINV` are the Excel 2007 names of `VAR.S`, `STDEV.S`, `T.DIST.2T`, `T.INV.2T` and `F.INV.RT`; they were chosen for compatibility and give identical results. The thresholds (0.30, 0.70, 0.60) and the confidence level are yellow input cells, so the decisions update automatically if the lecturer's criteria differ.